Analysis of Zoning Utilization Ratios in Newly Completed New York City Residential Buildings from 2012 to 2025.

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Project Background

New York City is facing a historic housing shortage with the 2025 vacancy rate declining to 1.4% according to the NYC Rent Guidelines Board[i]. A side effect of this historically low vacancy rate is rapidly increasing residential rents, with the median Manhattan rent reaching $4,750 in 2025 according to the real estate firm Douglas Elliman[ii]. The historic shortage of new residential homes in New York City is partly brought on by high development, land acquisition and regulatory costs for residential real estate developers which restricts the construction of new residential housing.

Once a property is acquired by a developer, they must determine what business strategy will bring them the greatest return on their investment. While the most efficient strategy would be to build as many units as the market demands, developers are restricted by the city on how large a structure they are able to build on a given lot. Because land costs in New York City are among the highest in the world, there is a business utility for developers to utilize 100% of their buildable floor area rights to maximize the number of sellable or rentable units, there is also a social utility to adding new high-quality homes for New York families to live in, reducing the demand burden on existing housing supply.

The density of a given project is restricted by which zone the plot of land is on. The city is divided into residential districts, each with rules dictating how large of a structure can be built on each plot in the district. The size of a building is formally dictated by land use regulators using the Floor Area Ratio (FAR). This indicator is calculated by measuring the floor area of the proposed structure divided by the area of the lot being developed. A zoning district having a FAR of 1 allows the developer to build a 1-story building covering the entire lot, a 2-story building covering half the lot, and so on. This is one of the most important metrics used to dictate land use regulations in the city.

Project Goals

The primary goal of this study is to identify the economic, regulatory, and physical factors that most significantly influence a developer's decision to maximize the allowable "Floor Area Ratio" (FAR) of the residential buildings they construct. By analyzing Certificates of Occupancy issued between July 2012 and November 2025, this project aims to explore the gap between legal FAR potential and built reality in the NYC housing market.

To achieve this, I will pursue the following objectives:

1. Determine Key Predictors of Maximum Utilization: I will employ a logistic regression model to quantify the effect of predictor variables such as land value, zoning district, lot irregularity, and landmark status on the probability of a building being "Maxed Out" (defined as a building with a zoning utilization ratio using greater than 90% of its allowable FAR).

2. Establish Statistical Robustness via Bootstrapping: To ensure the reliability of the model's findings, I will use non-parametric bootstrapping techniques to generate 95% confidence intervals for the logistic regression coefficients. This will allow for a more rigorous assessment of which variables have a statistically significant impact on development intensity.

3. Visualize Temporal Regimes: I will investigate how development intensity has shifted over time in response to industry shocks (such as the 2016 Zoning for Quality and Affordability amendments, the COVID-19 pandemic, and the 2022 expiration of the 421-a tax exemption). This will be achieved by applying a General Additive Model (GAM) to time series data of the certificates of occupancy issuance dates against the zoning utilization ratio of the buildings these certificates were issued to. This will allow for the visualization of non-linear trends and structural breaks in the data. Note: In my proposal I indicate that I would do this using a LOESS Locally Estimated Scatterplot Smoothing technique but because I am using a large dataset with over 200 thousand observations this data was not able to run on my computer or in a high RAM Google Colab environment. I instead used the General Additive Model because it is less computationally expensive while providing similar results.

Description of Data

I sourced the data for this study from four datasets obtained from NYC Open Data, a repository for public datasets managed by the NYC Office of Technology and Innovation, the four datasets are described below.

- The DOB Certificate of Occupancy data set is collection of all Certificates of Occupancy issued by the City of New York from July 2012 to March 2021; a Certificate of Occupancy states a building's legal use and/or type of permitted occupancy and is issued to new buildings once the NYC Department of Buildings inspects the completed building and deems it to be habitable.
- The DOB NOW Certificate of Occupancy is a collection of all Certificates of Occupancy issued by the City of New York from March 2021 to November 2025, this data is in a separate database due to the migration of department of building certificate of occupancy data from DOB Building Information System to DOB NOW in March of 2021.
- The DOB JOB Application Filings includes a count of the proposed dwelling units in each building in the previous datasets. The proposed dwelling unit variable is necessary for data cleaning purposes as I will filter out all nonresidential buildings from the certificate of occupancy dataset by excluding all buildings that received a final certificate of occupancy with less than 1 residential unit, the excluded buildings are typically non-residential structures such as garages and commercial buildings which are not covered under the scope of this study.
- The Primary Land Use Tax Lot Output (PLUTO) data set is where I sourced the Built FAR and Allowable FAR datapoints from. I will join this data to the combined certificate of occupancy dataset using the concatenated Borough, Block and Lot numbers, which serves as a unique identifier of each tax lot in New York City.

I combined these 4 datasets into a single project dataset titled (final_dataset) with 233 thousand observations and 15 variables. The variable names, outcomes, converted values and, type are stated in the below data card.

Variable Name	Outcomes	Converted Values	Type
Job_Number	Unique identifiers	Original Values (Character string)	Categorical (Nominal)
Issue_Date	Date values (YYYY-MM-DD)	Original Values	Date / Time
FAR_Utilization	Continuous numeric ratio	Calculated (BuiltFAR / AllowableFAR)	Numerical (Continuous)
Maxed_Out	0, 1	1 (Yes, Ratio > 0.9), 0 (No)	Categorical (Binary)
BuiltFAR	Numeric values	Original Values (Cleaned to numeric)	Numerical (Continuous)
AllowableFAR	Numeric values	Original Values (Cleaned to numeric)	Numerical (Continuous)
Borough	Boroughs: Bronx, Brooklyn, Manhattan, Queens, Staten Island	Original Values (Title Cased)	Categorical (Nominal)
Zoning_Category	R, C, M	Factor levels (First letter of zone)	Categorical (Nominal)
Is_Landmark	No, Yes	Factor levels ("No", "Yes")	Categorical (Binary)
Is_Irregular	false, true	Factor levels ("false", "true")	Categorical (Binary)
LotFront	Numeric values (Linear feet)	Original Values	Numerical (Continuous)
Land_Value_Per_SqFt	Numeric values (USD $)	Calculated (AssessLand / LotSize)	Numerical (Continuous)
Job_Type	New Building, Alteration 1	Original Values	Categorical (Nominal)
Dwelling_Units	Numeric count (Integers ≥ 1)	Original Values	Numerical (Discrete)
LotSize	Numeric values (Square feet)	Original Values	Numerical (Continuous)

Response Variables

I created continuous and binary response variables to gain insights about the effect different variables have on a developer's decision to use as much of the floor area ratio as legally allowed. The zoning utilization ratio (FAR_Utiilization) was created by dividing the Built FAR (Built_FAR) by the Allowable FAR (Allowable_FAR). The maxed out (Maxed_Out) binary variable was created by dividing the zoning utilization ratio into groups where a building was maxed out (FAR_Utiilization > .9) versus not maxed out (FAR_Utiilization < .9).

Exploratory Data Analysis

Correlation Chart of Continuous Variables

	FAR_Utilization	Land_Value	Lot_Frontage	Lot_Size	Dwelling_Units
FAR_Utilization	1	0.2595	0.0012	-0.0917	0.3632
Land_Value	0.2595	1	0.0107	-0.0448	0.4288
Lot_Frontage	0.0012	0.0107	1	0.8747	0.2281
Lot_Size	-0.0917	-0.0448	0.8747	1	0.0232
Dwelling_Units	0.3632	0.4288	0.2281	0.0232	1

Findings:

- Land value has a positive but relatively weak correlation with the zoning utilization ratio at (.2595). While higher land costs do encourage density, this weak correlation suggests that price alone is not the sole predictor. Regulatory or physical constraints likely play a significant role in the decision to max out.
- Lot frontage and lot size are highly corelated (.8747) which is to be expected since they are both measurements for the size of lots. If I were to include both variables in my logistic regression model it would introduce a risk of collinearity. Because of this I will drop lot frontage and keep lot size because lot size is a better measure of development potential than lot frontage.
- The zoning utilization ratio and lot size show a negligible negative correlation of (-.0917). This suggests that the total area of a lot is not a determinant of whether a developer maximizes their allowable floor area, implying that zoning utilization rates remain consistent regardless of whether the project is on a small or large lot.

Pearson Chi-Squared Analysis of Categorical Predictors

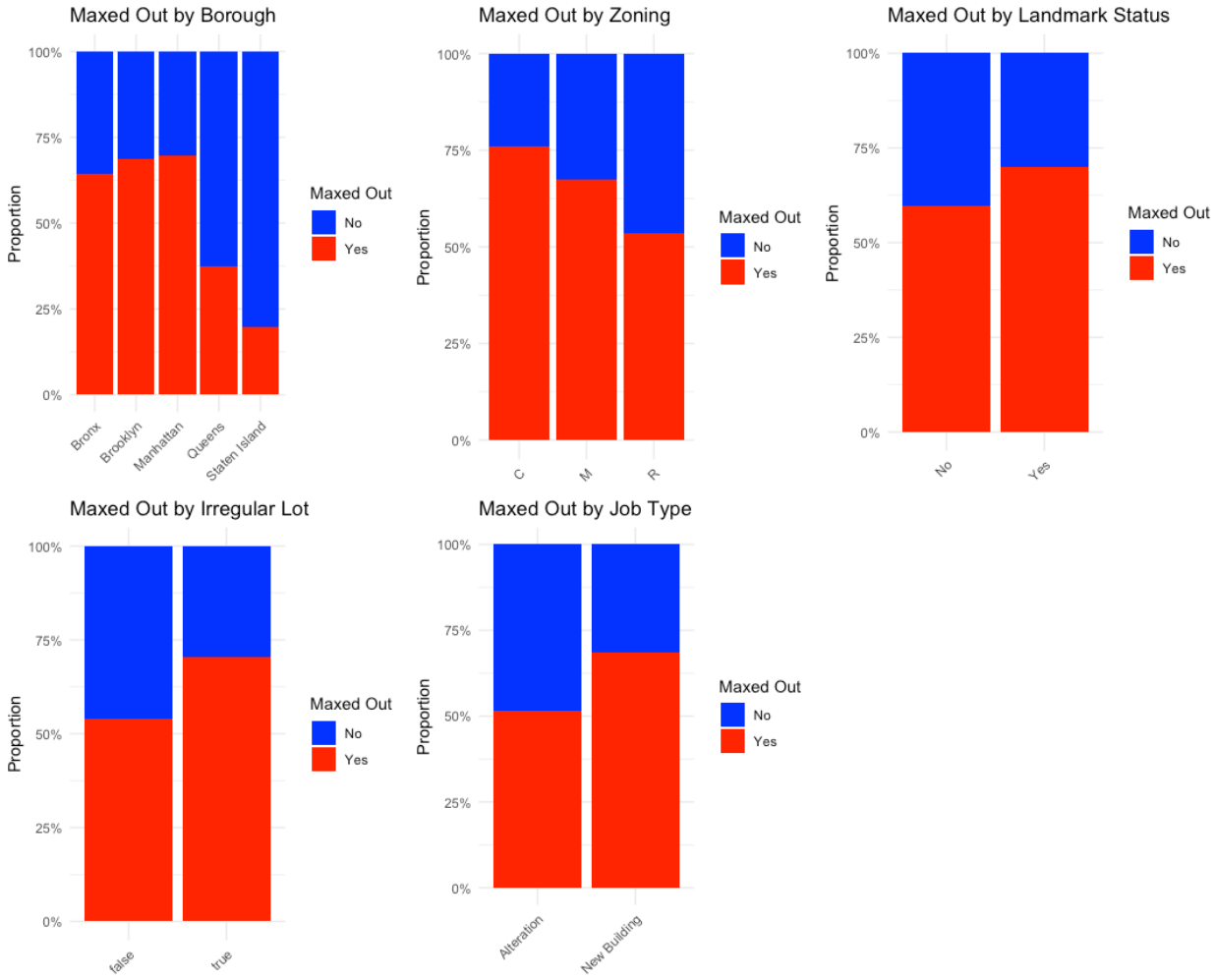
Variable	X-squared Statistic	Degrees of Freedom (df)	P-Value
Borough	24,157	4	< 0.001
Zoning Category	9,094.40	2	< 0.001
Irregular Lot	5,901.70	1	< 0.001
Landmark Status	211.5	1	< 0.001

Findings:

- Project borough shows a strong statistical dependency with our binary response variable maxed out. The X-squared value of 24,157 indicates that the borough a project is located in is a strong predictor of a zoning utilization ratio above .9.
- Zoning category and our binary response variable maxed out also show a strong dependency with an X-squared value of 9,094. This confirms that the difference between Residential, Commercial, and Manufacturing zones is an important driver of development decisions.

- Is irregular and our binary response variable maxed out produced an X-squared value of 5,901, confirming that lot geometry is a major factor in development. This aligns with the regression findings that irregular lots are utilized differently than standard rectangular lots.
- Is landmark and our binary response variable maxed out are statistically associated, though the relationship is weaker than other factors with an X-squared value of 211.5. While landmark status is a significant constraint, this lower statistic suggests that it is a less dominant predictor of zoning utilization.

Visual Analysis of Categorical Associations



Findings:

Based on the relative proportions of "Yes" (Maxed Out) and "No" (Not Maxed Out) categories the higher utilization (Maxed Out) groups are:

- Boroughs with high land value: Manhattan and Brooklyn show the highest proportions of properties maxing out their allowable floor area.

- New Construction: Projects categorized as new building job types are significantly more likely to max out than alteration job types (projects that add additional square footage to existing structures).
- Commercial/Mixed Zoning: Properties in commercial (C) or mixed-use zoning districts tend to have higher utilization rates compared to pure manufacturing (M) districts.
- Irregular Lots: Properties with lots deemed as irregularly shaped by the Department of Finance (often corner lots) show a surprisingly high rate of maximizing zoning. I was surprised by this outcome, so I investigated this further. It turns out the department of finance deems all corner lots as irregular. The result makes sense because in a street of row houses the corner lots are often the most desirable since units in these buildings can have better exposure to sunlight.

The lower utilization (Not Maxed Out) groups are:

- Lower Density Boroughs: Staten Island and Queens show a much lower proportion of properties maxing out their zoning rights.
- Alterations: Permits filed as "Alterations" result in maxing out the full zoning capacity much less frequently compared to new developments. This is possibly because it is difficult to add significantly more floor area to a building with what is often a much older foundation built during a more restrictive zoning regime.
- Manufacturing Zones: Properties in Manufacturing (M) zones appear less likely to maximize residential floor area ratios compared to mixed use and residential only zones.

Neutral or Weaker impact can be found as the following:

- Landmark Status: While Landmarked properties do max out less compared to non-landmarked properties, the statistical association (Chi-Squared) was weaker than for Borough or Zoning, suggesting that location is a stronger driver than historic status alone.

Logistic Regression Model

The logistic regression model was specified to quantify the probability of a building maximizing its zoning rights. The predictor variables were selected to capture economic incentives (Land Value per Square Foot), physical constraints (Lot Size, Irregular Lot Status), and regulatory jurisdictions (Borough, Zoning Category, Landmark Status). Lot frontage was excluded in favor of lot size to avoid collinearity, as stated in the section on correlation analysis. Additionally, job type was included to control for the observed disparity in utilization rates between new construction and alteration projects, ensuring the model accounts for the differing scopes of development.

```
glm(formula = Maxed_Out ~ Borough + Zoning_Category + Is_Landmark +
    Is_Irregular + Job_Type + LotSize + Land_Value_Per_SqFt,
    family = "binomial", data = final_dataset)

Coefficients:
                      Estimate Std. Error z value Pr(>|z|)
(Intercept)         -2.046e-01  2.305e-02  -8.876   <2e-16 ***
BoroughBrooklyn      4.365e-01  1.823e-02  23.943   <2e-16 ***
BoroughManhattan     3.937e-02  2.031e-02   1.939   0.0525 .
BoroughQueens       -9.506e-01  1.885e-02 -50.434   <2e-16 ***
BoroughStaten Island -1.951e+00  2.731e-02 -71.450   <2e-16 ***
Zoning_CategoryM    -1.959e-01  2.150e-02  -9.110   <2e-16 ***
Zoning_CategoryR    -2.423e-01  1.465e-02 -16.542   <2e-16 ***
Is_LandmarkYes      -3.750e-03  3.362e-02  -0.112   0.9112
Is_Irregulartrue     5.710e-01  1.046e-02  54.614   <2e-16 ***
Job_TypeNew Building 1.075e+00  1.034e-02 103.949   <2e-16 ***
LotSize             -1.051e-07  6.284e-09 -16.729   <2e-16 ***
Land_Value_Per_SqFt  2.913e-03  5.169e-05  56.352   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 315086  on 233989  degrees of freedom
Residual deviance: 266330  on 233978  degrees of freedom
AIC: 266354

Number of Fisher Scoring iterations: 5
```

Odds Ratio

```
exp(coef(logit_model))
      (Intercept)      BoroughBrooklyn     BoroughManhattan        BoroughQueens BoroughStaten Island
        0.8149963            1.5472745           1.0401555            0.3864922            0.1420707
  Zoning_CategoryM     Zoning_CategoryR      Is_LandmarkYes     Is_Irregulartrue Job_TypeNew Building
        0.8221162            0.7848204           0.9962565            1.7701238            2.9306611
       LotSize  Land_Value_Per_SqFt
        0.9999999            1.0029171
```

Findings:

- Based on p-values, the important features are Borough (Brooklyn, Queens, Staten Island), Zoning_Category (Manufacturing, Residential), Is_Irregular, Job_Type (New Building), LotSize, and Land_Value_Per_SqFt, as they have p-values less than 0.05.
- The features that failed to meet significance are Is_Landmark (p=0.9112) and Borough (Manhattan) (p=0.0525) as they have p-values greater than 0.05. A project being in a landmark district compared to a non-landmarked district has little effect on whether a project will max out. The same is true for a building that is in Manhattan versus our baseline borough (The Bronx)
- The null deviance is 315,086, this is the deviance of the model with the intercept and no predictors included. The residual deviance is 266,330, this is the deviance of the model with all the predictors included. This decrease in deviance reflects the significant improvement of the model once all the predictors are included.
- The coefficient with the greatest positive effect on the probability of maxing out is the Job_Type (New Building) coefficient at (1.075). Per my odds ratio calculation a new building job type is almost 3 times more likely to max out than alteration job type, indicating that new construction projects are significantly more likely to utilize their full zoning rights compared to alterations of existing buildings. The coefficient with the greatest negative effect is Borough (Staten Island) (-1.951). Per our odds ratio, a project in Staten Island has only a 15% chance of maxing out compared to our baseline borough (The Bronx).

Multi-Collinearity Test (VIF)

```
> vif(logit_model)
                      GVIF Df GVIF^(1/(2*Df))
Borough           2.038478  4        1.093108
Zoning_Category   1.568661  2        1.119134
Is_Landmark       1.032331  1        1.016037
Is_Irregular      1.077848  1        1.038195
Job_Type          1.199597  1        1.095261
LotSize           1.045503  1        1.022498
Land_Value_Per_SqFt 1.787964  1        1.337148
```

Findings: To ensure the stability of the regression estimates, a Variance Inflation Factor (VIF) test was conducted. All predictors yielded VIF values below 2.1 (well below the standard threshold of 5.0), confirming that multicollinearity is not present in the model.

Bootstrapping of Logistic Regression Model Coefficients

To gauge the reliability of the logistic regression estimates and to test for potential overfitting, I conducted a non-parametric bootstrap analysis of the 95% confidence intervals using 100 resamples. The size of the resamples was kept to 100 due to the large size of my dataset.

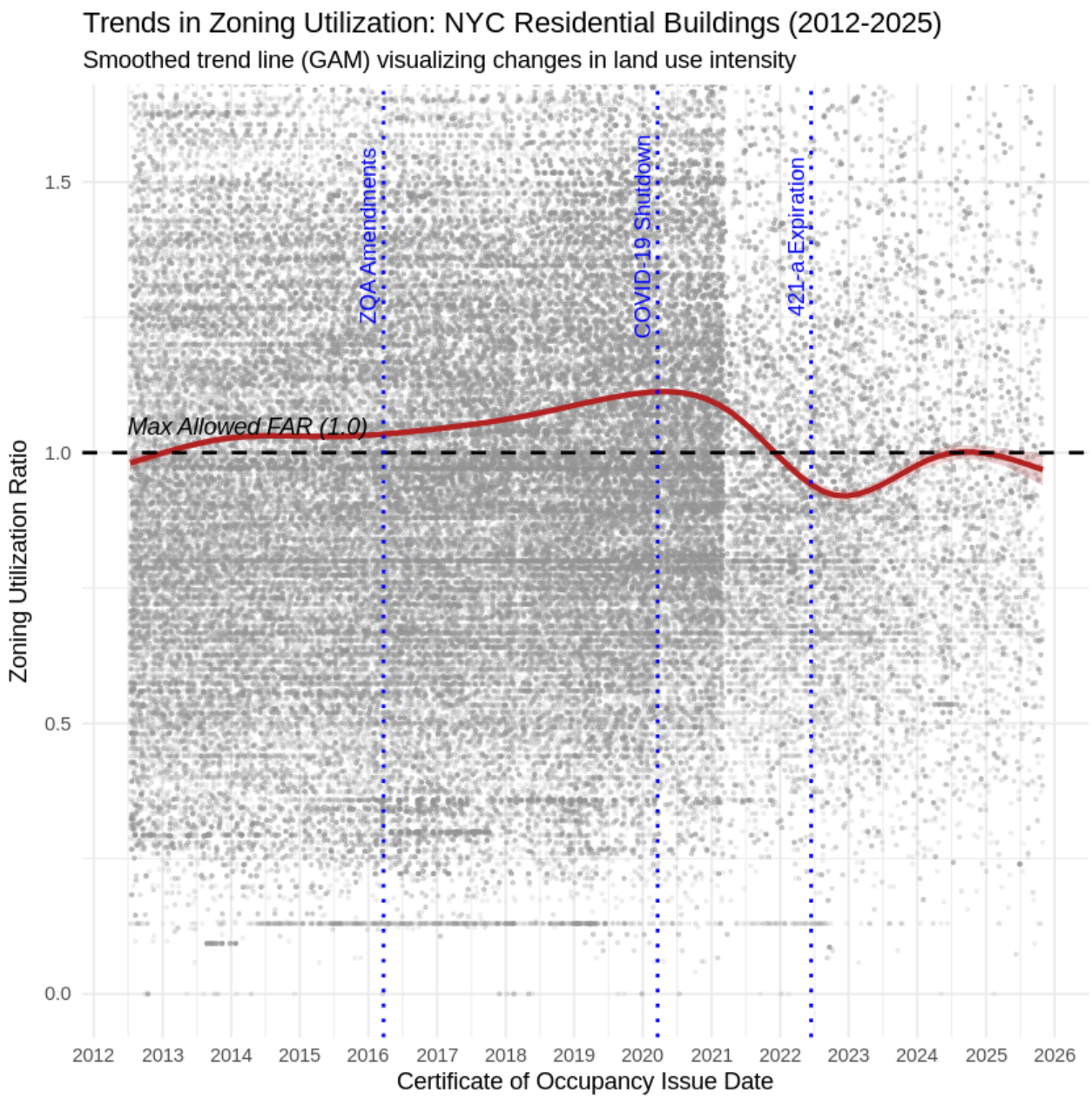
Bootstrap ID	Variable Name	Original Coeff	Std. Error	95% CI (Lower)	95% CI (Upper)
t1*	(Intercept)	-0.2046	0.0222	-0.2481	-0.1611
t2*	Borough: Brooklyn	0.4365	0.0186	0.4	0.473
t3*	Borough: Manhattan	0.0394	0.0213	-0.0023	0.0811
t4*	Borough: Queens	-0.9506	0.0194	-0.9886	-0.9126
t5*	Borough: Staten Is.	-1.9514	0.0247	-1.9998	-1.903
t6*	Zoning: Manufacturing	-0.1959	0.0247	-0.2443	-0.1475
t7*	Zoning: Residential	-0.2423	0.0141	-0.2699	-0.2147
t8*	Is_Landmark	-0.0038	0.0363	-0.0749	0.0673
t9*	Is_Irregular	0.571	0.01	0.5514	0.5906
t10*	Job_Type: New Bldg	1.0752	0.0112	1.0532	1.0972
t11*	Lot Size	-1.05E-07	4.80E-09	-1.14E-07	-9.60E-08
t12*	Land Value Per SqFt	0.0029	0.0001	0.0028	0.003

The bootstrapped 95% confidence intervals of the logistic regression coefficients confirm the results of our exploratory data analysis and odds ratio analysis. The bootstrapped confidence intervals for Job Type (New Building), Borough (Staten Island), and Lot Shape (Is Irregular) do not include 0. We can thus conclude that they are significant predictors that a building will or will not be maxed out. The bootstrapped 95% confidence intervals also confirm that predictors Borough (Manhattan) and Landmark Status (Is Landmark) are not significant predictors seeing as the 95% confidence interval for these coefficients both include 0. This means that we cannot say with 95% confidence that these predictors are statistically significant.

Temporal Analysis of Zoning Utilization

To show how development intensity has changed over time in response to events with significant impacts to housing construction in New York City, a General Additive Model (GAM) smoothing curve was applied to the time-series data of the Certificates of Occupancy plotted against the zoning utilization ratio between 2012 and 2025. Using GAM, we can visualize trends in the zoning utilization ratio without assuming a linear pattern.

Our GAM trendline shows a slow but steady increase from 2012-2019 with a faster increase occurring from 2019 and then peaking in 2021. This can be explained by a resurgence in market demand after the great recession and the lagged effects that the Zoning for Quality and Affordability Act (2016) had on the developer's ability to increase unit counts and building height to maximize the allowed floor area ratio. The Zoning for Quality and Affordability Act increased maximum building height and maximum first floor heights[iii]. These changes allowed developers in mixed use districts to build first floor retail on their properties and build taller buildings in exchange for affordable units reserved for seniors or low-income residents which allowed them to take full advantage of their allowed FAR.



The rise in zoning utilization peaks shortly after the start of the March 2020 Covid 19 lockdown in New York State. After this peak the trendline experiences a sharp declining phase where the average utilization ratio dropped significantly, falling below the maximum allowable baseline (1.0) and bottoming out near 0.90 in 2022. This can be explained by the effects of economic uncertainty, the collapsing of the rental market and the increase in construction and labor costs. This suggests developers may have lowered their initial ambitions to maximize the zoning utilization ratios for the residential projects they were constructing.

The June 2022 expiration of the 421-a tax exemption, a total tax exemption for buildings that set aside a portion of their units to income restricted residents, led to a lagged rise in the zoning utilization ratio followed by lagged decline. I hypothesis that this is because developers rushed to start construction of residential buildings that maximized unit counts to provide the highest number of market rate units because the deadline was a construction start deadline. The timing of the increase and decline is subject to a lag as new residential construction projects in New York typically take from 1 to 3 years to complete after they are commenced. This explains why zoning utilization reaches a local peak in mid 2024, two years after the 421-a tax exemption construction start deadline.

Final Analysis and Recommendations to City Policy Makers

After running a logistic regression, bootstrap validation, and a temporal GAM analysis, I determined that the zoning utilization ratio of buildings built in New York is dependent on economic, regulatory, and physical factors and is not determined solely by land price. Our analysis names the following factors as the most likely to increase the probability that a project is maxed out.

- Construction Type: With a log ratio coefficient of 1.075 (Odds Ratio: 2.93), the decision to construct a "New Building" over an “Alteration” significantly increases the probability that a new building is built to its legal limits. This suggests that there may be significant physical and regulatory constraints preventing altered projects from building to their legal limits.

- Project location by Borough (Brooklyn, Queens and Staten Island): While Brooklyn has a high zoning area utilization compared to The Bronx (log coefficient of .427, odds ratio of 1.55), Queens (log coef -.951, odds ratio .386) and Staten Island (log coef -1.951, odds ratio .142) have very low zoning utilization ratios suggesting that new construction projects in low density boroughs are more likely to maintain the character of the neighborhoods they are constructing in rather than maximize their legally allowed floor area ratios.

- Effect of industry shocks: Our GAM temporal analysis showed how changes in policy (ZQA Act and expiration of the 421-a Tax Incentive) and exogenous events (COVID-19 pandemic) can affect zoning utilization over time. While the ZQA Act increased zoning utilization ratios the uncertainty and negative economic impacts of the COVID 19 pandemic greatly reduced the ability or willingness of developers to fully utilize their lots. The expiration of the 421-a tax benefits had a more mixed effect with a lagged increase and lagged decrease following the construction start deadline of the expired program.

If the goal of city leaders is to encourage the private construction of new residential housing, there are many tools at their disposal that will help them achieve this goal. Helping developers maximize the floor area ratio of residential buildings is a way to work towards this goal without needing to pass any new zoning reforms. After my analysis I recommend the following:

- Make it easier to demolish existing buildings: Our logistic regression results indicate that developers are disincentivized from maxing out the floor area of projects that are altered rather than newly constructed, a possible solution to this would be to make it

easier to tear down these buildings so that the developers are able to construct new buildings in their place rather than alter them

- Incentivize increased density in Staten Island and Queens: The under building of units in Staten Island and Queens represents a missed opportunity for solving the housing crisis since these areas are home to much of the unutilized and underutilized residential land in the city. Since developers in these areas voluntarily build below the legal limit, the Department of City Planning should investigate whether this is due to restrictive bulk regulations (such as height limits or parking requirements) or weak market economics. Density bonuses and reduced parking mandates in these Boroughs should be introduced to get developers to fully utilize their residential lots.

- Increase tax incentives for building affordable units: The state legislature is ahead of me on this front with the passage of the 485x tax incentive last year that is meant to replace the expired 421a benefit. Our GAM model shows that the expiration of the 421a benefit led to a decline in zoning utilization after a rush to qualify for the program. What I gather from this is that this incentive is very valuable to developers and influences the zoning utilization ratios of the residential buildings that they construct. The state legislature must not let these benefits expire without a suitable replacement in the pipeline.

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[i] Rent Guidelines Board: Housing Supply Report

[ii] Douglass Elliman median Manhattan rents

[iii] Department of City Planning: Zoning for Quality and Affordability